

An elastic network model with $k(l)$: building it, and what it does

2026-08-25. Acylphosphatase, 2acy chain A: 98 residues, 960 contacts at $d_{\max} = 10.5$ Å, uniform $k_{ij} = 1$ (ANM). Everything seeded and reproducible from analyses/.

1. What this is

An elastic network model in which the spring constants depend on the rest lengths, $k_{ij}(l_{ij})$, rather than on the equilibrium distances. The rest lengths are then the only parameters, and the contact map follows from them rather than from the structure.

Mutants are made by perturbing rest lengths and **minimising the energy numerically**. No linear response and no expansion of any kind enters the dynamics; the quadratic expansion is used only to look at states already accepted.

What follows is the model and five things measured with it:

§2	the model: potential, state, mutation, energy differences
§3	trajectories — do they run, and what do they do
§4	scans — site profiles of energy, structure and entropy, with $k(l)$ live and frozen
§5	contact-map statistics — how often, and in which direction, mutation rewires the network
§6	refitting a strained structure — does discarding frustration change the motions
§7	what these say together

§6 is the longest section because it needed two controls before its answer could be trusted, not because it is the most important.

Results in brief

1. **Trajectories run.** No stalling at any selection strength tried, and they keep producing strained structures indefinitely (§3). The reason they do not stall is that strain *widens* the distribution of ΔV without shifting its mean: a strained network survives on its left tail, not by drifting downhill.

2. **Energy profiles barely notice $k(l)$; entropy profiles do.** Site profiles of the energy cost of mutation are almost unchanged when k is allowed to follow l (correlation 0.974), and reproduce the standard results — buried sites cost more to mutate and move less. The range of $\Delta(TS)$ across sites widens 6.25-fold (§4). Adding or removing a spring changes the spectrum far more than moving a rest length does.
3. **The contact map moves constantly.** At the working mutation size, three-quarters of mutations change it, with gains and losses balanced, so the network rewires rather than erodes (§5).
4. **Frustration can be neglected.** Refitting an ENM to a strained structure — standard practice — preserves the RMSF profile and the soft-mode subspace, and does so far better than a random perturbation of the same size would (§6).
5. **But most of what refitting changes is not frustration.** It is the contact map being re-derived. Under a hard cutoff that term is $5.5\times$ the frustration term; under a soft one the two are comparable. Which dominates is a modelling choice, usually made without comment (§6).

2. A model whose parameters are the rest lengths

2.1 The potential

$$V(\mathbf{r}; \{l_{ij}\}) = V_0 + \frac{1}{2} \sum_{i < j} k_{ij}(l_{ij}) (d_{ij}(\mathbf{r}) - l_{ij})^2$$

The rest lengths are the only parameters. The single departure from a standard ENM is that the spring constant is a function of the **rest length**, $k_{ij}(l_{ij})$, not of the equilibrium distance $k_{ij}(d_{ij}^e)$.

At the wild type these are the same function, since the fit sets $l_{ij} = d_{ij}^e$. They differ only for a mutant, and then in a way that matters:

the contact map becomes a property of the parameters, not of the structure.

Contacts are made and broken by *mutation*, when an l_{ij} crosses the cutoff — not by relaxation. For the ANM, $k_{ij}(l_{ij}) = \mathbb{1}[l_{ij} \leq d_{\max}]$, with backbone neighbours always bonded. A pair with $k_{ij} > 0$ is **active**; the active pairs are the contact map. $V_0 = 0$ throughout, so V of a state is exactly its strain energy.

2.2 The state

A state carries l_{ij} for **all** $N(N-1)/2 = 4753$ pairs, not just the 960 active ones, initialised to $l_{ij} = d_{ij}(\mathbf{r}_{\text{wt}}^e)$.

This is forced, not convenient. Every *active* wild-type pair has $l_{ij} \leq d_{\max}$ by construction, so if only active pairs carried a rest length no inactive pair could

ever be pulled inside the cutoff, and the network could only thin. Carrying all pairs makes breaking and forming symmetric. The cost is negligible: a force evaluation over 4753 pairs takes ~ 4 ms against 30 ms for one diagonalisation, and the Hessian stays $3N \times 3N$ because only active pairs contribute.

2.3 One mutation

Pick a site j ; perturb the rest length of every pair involving j by $\delta l_{ij} \sim \mathcal{N}(0, \sigma^2)$; recompute $k_{ij}(l_{ij})$; **minimise V numerically** for $\mathbf{r}_{\text{mut}}^e$; evaluate V exactly there. $\sigma = 0.3 \text{ \AA}$ unless stated.

2.4 The discipline

No expansion ever conditions the dynamics. The structure comes from minimising V — not linear response, not $\mathbf{C}_{\text{wt}}\mathbf{f}$, not a two-term formula. Energies are read at the minimum. Normal modes are computed only to *observe* a state already accepted.

This is affordable: one exact minimisation converges in 4–20 Newton steps at ~ 0.25 s, so a trajectory runs at ~ 4 proposals per second. There is no performance argument for the linear-response shortcut.

2.5 Energy differences

The reference is always named, because along a trajectory it is a strained state, not the founder:

$$\begin{aligned}\Delta V_{\text{stress}} &= V_{\text{mut}}(\mathbf{r}_{\text{ref}}) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}) \\ \Delta V_{\text{relax}} &= V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{mut}}(\mathbf{r}_{\text{ref}}) \\ \Delta V_{\text{min}} &= V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}) = \Delta V_{\text{stress}} + \Delta V_{\text{relax}}\end{aligned}$$

Two Hamiltonians at one structure; one Hamiltonian at two structures; the difference of minima, which is what a trajectory accepts on.

All three are computed by **evaluating Hamiltonians**, never by a closed form. Here is why that matters. Write $l_{ij}^{\text{mut}} = l_{ij}^{\text{ref}} + \delta l_{ij}$, $k_{ij}^{\text{mut}} = k_{ij}^{\text{ref}} + \delta k_{ij}$, and let $r_{ij} \equiv d_{ij}(\mathbf{r}_{\text{ref}}) - l_{ij}^{\text{ref}}$ be the reference's own residual. Then

$$\Delta V_{\text{stress}} = \underbrace{\frac{1}{2} \sum_{ij} k_{ij}^{\text{ref}} \delta l_{ij}^2}_{\text{(a) always present}} - \underbrace{\sum_{ij} k_{ij}^{\text{ref}} r_{ij} \delta l_{ij}}_{\text{(b) zero only if the reference is relaxed}} + \underbrace{\frac{1}{2} \sum_{ij} \delta k_{ij} (r_{ij} - \delta l_{ij})^2}_{\text{(c) zero only if } \delta k_{ij}=0}$$

Term (b) is the **energy cross term** — a different object from the Hessian's transverse term g_{ij} below, though both vanish at a relaxed reference. It is

proportional to the reference's existing strain, it is the only term that can be negative, and it disappears at the founder. That is why a formula derived at the wild type appears to work and then fails along a trajectory.

Term (c) exists only in this model. The LFENM (Linearly Forced ENM, the model `penm` implements) assumes $\delta k_{ij} = 0$; here k follows l , so a spring crossing the cutoff contributes its whole stored energy discontinuously.

Verified numerically (`checks/test_core.R`), one mutation at site 40, $\sigma = 0.3$: at the relaxed founder (a) = +0.8516, (b) = 0 exactly, (c) = -0.0322; at a reference 15 substitutions along, (a) = +0.8516, (b) = -0.0314, (c) = +0.0555. Both reproduce the directly evaluated ΔV_{stress} to 10^{-15} .

Evaluating the Hamiltonians costs the same and cannot drop a term.

Dropping the Δ is not a notational nicety. Fifteen substitutions along, $V_{\text{mut}}(\mathbf{r}_{\text{ref}}) = 7.90$ while $\Delta V_{\text{stress}} = 1.06$ — a sevenfold difference, exactly V_{ref} , growing with the walk.

2.6 Definitions

Strain, two measures: $s_{\text{max}} = \max_{ij} |d_{ij}(\mathbf{r}^e) - l_{ij}|$ in Å, and $E_{\text{strain}} = \frac{1}{2} \sum k_{ij} (d_{ij}(\mathbf{r}^e) - l_{ij})^2$, equal to V since $V_0 = 0$.

Frustration: no conformation relaxes every spring at once, i.e. $E_{\text{strain}} > 0$ at the minimum.

Hessian, for an active pair,

$$\mathbf{K}_{ij} = -k_{ij} \left[\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T + g_{ij} (\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T - \mathbf{I}) \right], \quad g_{ij} = \frac{l_{ij}}{d_{ij}(\mathbf{r})} - 1$$

with $\mathbf{K}_{ii} = -\sum_{j \neq i} \mathbf{K}_{ij}$. The g_{ij} piece is the **transverse term**; it vanishes exactly when a spring is at its rest length, so **frustration reaches the dynamics only through g_{ij}** . Standard ENMs drop it because for them it is identically zero.

RMSF of site i : $\sqrt{\langle |\delta \mathbf{r}_i|^2 \rangle}$, from the 3×3 diagonal block of $\mathbf{C} = \beta^{-1} \mathbf{K}^+$.

RMSIP over $M = 20$ modes: $\sqrt{M^{-1} \sum_{a,b} (\mathbf{u}_a^A \cdot \mathbf{u}_b^B)^2}$. It scores a *subspace*.

Mode matching is a one-to-one assignment maximising total overlap (`clue::solve_LSAP`), not row-wise maxima, which are not a matching and inflate overlaps.

$\beta = 1$ wherever entropy appears — with $k_{ij} = 1$ this is a convention, not a physical temperature.

3. Trajectories

Question. Does a walk in this model run at all — and if it does, what keeps it running when every mutation is uphill from a relaxed network?

What I did. From the wild type, repeatedly: draw a site uniformly, mutate it (§2.3), minimise exactly, accept with probability $\min\{1, e^{-\nu\Delta V_{\min}}\}$ measured against the *current* state. A neutral walk sets $\nu = 0$ and accepts everything. 120 substitutions at three selection strengths, seeds $300 + 10\nu$.

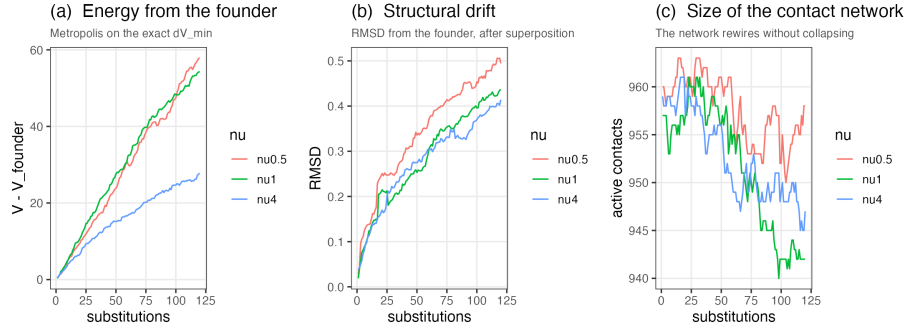


Figure 1. Three walks at $\nu = 0.5, 1, 4$ (colour), 120 accepted substitutions each, $\sigma = 0.3$. x is accepted substitutions, not proposals. **(a)** V relative to the founder — since $V_0 = 0$, the state’s strain energy. **(b)** RMSD from the founder after rigid-body superposition, Å. **(c)** Number of active pairs; the wild type has 960.

Result.

ν	final E_{strain}	final s_{max} (Å)	active pairs	acceptance
0.5	57.85	1.479	958	0.727
1	54.26	1.307	942	0.612
4	27.76	0.977	947	0.198

No run stalled. Selection sets the energy level; the network ends within 2 % of its starting size in every case.

What it means. The walks produce genuinely strained states — s_{max} of 1–1.5 Å against a 10.5 Å cutoff — and keep producing them indefinitely.

An earlier exploration (`sclfenm/MODEL.md` §6) proves that from a *relaxed* network, $\Delta V_{\min} \geq 0$ always. Its hypothesis fails here after the first mutation, so the question is empirical.

What I did. One neutral walk, 40 substitutions, seed 2, saving every 10. From each saved state (and from the wild type), 200 fresh trial mutations, each minimised exactly, $\Delta V_{\min}$ recorded and the trial discarded.

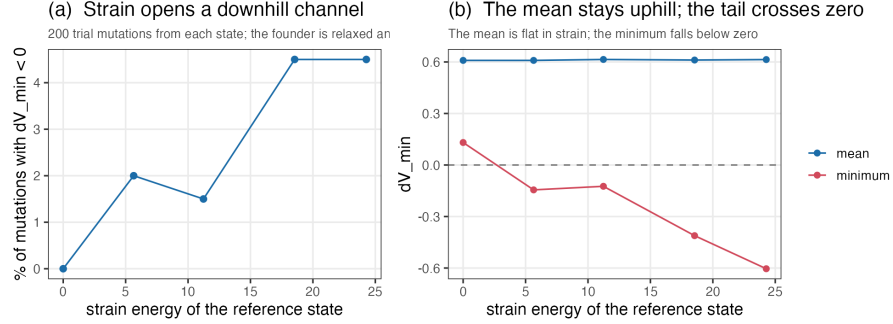


Figure 2. x is E_{strain} of the reference state. **(a)** Percent of the 200 trials with $\Delta V_{\text{min}} < 0$. **(b)** Mean (blue) and minimum (red) of the same 200 values.

substitutions	E_{strain}	mean ΔV_{min}	min	fraction < 0
0 (founder)	0.00	+0.609	+0.131	0.000
20	11.23	+0.615	-0.124	0.015
40	24.28	+0.614	-0.604	0.045

From the relaxed founder not one of 200 mutations is downhill; from a strained state a few per cent are. The mean is unmoved at ~ 0.61 throughout: **strain widens the distribution rather than shifting it**, so the walk keeps finding acceptable moves without drifting downhill.

4. Scans: site profiles under a single mutation

What I did. Mutate each of the 98 sites 8 times from the fixed wild type ($\sigma = 0.3$, seed 11), minimise each exactly, discard, average per site. Repeat with k_{ij} frozen at wild-type values — same seed, hence the same perturbations — so the two differ only through $k(l)$.

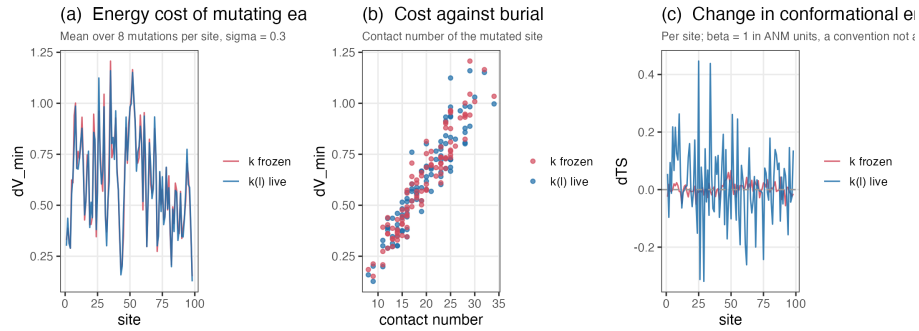


Figure 3. Blue = $k_{ij}(l_{ij})$ live, red = frozen. **(a)** $\Delta V_{\min}$ per site. **(b)** The same against the mutated site’s contact number. **(c)** $\Delta(TS)$ per site, $\beta = 1$.

(over the 98 sites)	$k_{ij}(l_{ij})$ live	k_{ij} frozen
$\text{cor}(\Delta V_{\min}, cn_j)$	+0.928	+0.946
$\text{cor}(\delta \mathbf{r}_j ^2, cn_j)$	−0.657	−0.712
range of $\Delta(TS)$	0.765	0.122

Both reproduce the standard results — buried sites cost more to mutate and move less. The energy profiles agree closely (correlation 0.974). The entropy range differs 6.25-fold: this is where the contact map shows up.

5. How the contact map moves

Question. If $k_{ij}(l_{ij})$ never changed the active set, the model would be an ordinary ENM with extra bookkeeping. How often does it change it, and does the network rewire or simply erode?

What I did. From the wild type, mutate one uniformly chosen site (§2.3), recompute k_{ij} , compare active sets. No minimisation — this is a property of the parameters alone. 200 draws at each σ , seed 1.

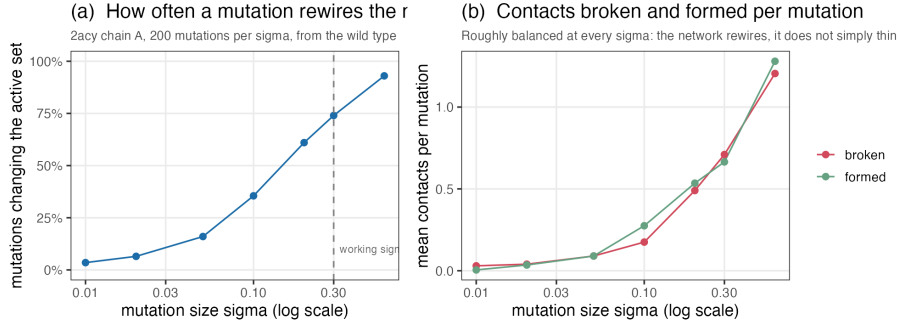


Figure 4. **(a)** Fraction of the 200 draws changing the active set, log x ; dashed line at the working $\sigma = 0.3$. **(b)** Mean pairs gained (green) and lost (red) per mutation.

σ	changing the active set	broken	formed
0.01	3.5 %	0.030	0.005
0.10	35.5 %	0.175	0.275
0.30	74 %	0.71	0.67
0.60	93 %	1.21	1.28

At the working σ , three-quarters of mutations rewire the network, and gains and losses are balanced — it rewires rather than erodes.

6. Refitting a strained structure

Question. A trajectory leaves you with a structure whose springs are strained. If you now do what everyone does — throw away the rest lengths, refit $l_{ij} = d_{ij}$ to the coordinates, diagonalise — do you get the same motions?

The two models. Both live at the **same coordinates** $\mathbf{r}^e$; they differ only in what they remember about how the structure got there.

- **Frustrated:** the state’s real parameters. Its springs are not at their rest lengths, so $g_{ij} \neq 0$ and the transverse term survives in $\mathbf{K}$. This is the truth, within the model.
- **Rebuilt:** forget the parameters and fit fresh springs to the coordinates. Then $l_{ij} = d_{ij}$ by construction, so $g_{ij} = 0$ and the transverse term vanishes. Its contact map is also re-derived from the coordinates rather than inherited. This is what an ENM practitioner does with a mutant structure.

The difference between them *is* the frustration the standard construction discards. Everything below measures that difference.

What I did. Three independent neutral walks of 60 substitutions (seeds 501–503), saving the state every 10 — **18 states**. At each state, build both models and compare their RMSF profiles, their 20 softest modes, and their entropies.

One precaution matters. Both models must sit at a true minimum before any spectrum is taken: off a stationary point the frustrated Hessian grows spurious near-zero and negative eigenvalues, and since $\text{RMSF} \sim 1/\lambda$ that ruins every number. Verified for all 18 states — exactly six zero modes, none negative.

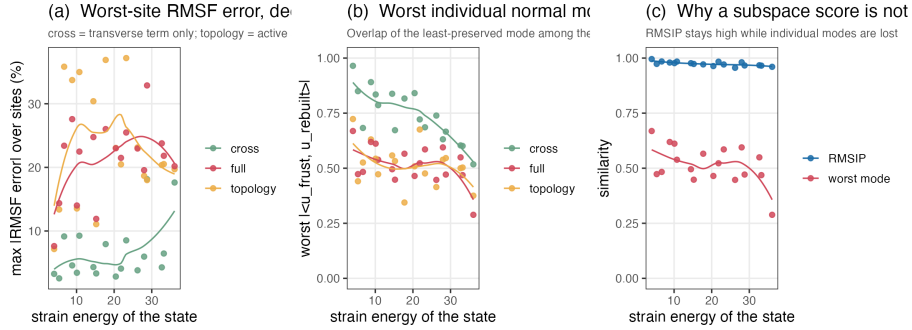


Figure 5. Rebuild error against accumulated strain, 18 states pooled from three walks (not 18 independent draws). x is E_{strain} . Colours decompose the difference: **green** = transverse term alone (same active set, $g_{ij} \neq 0$ vs $g_{ij} = 0$); **yellow** = active set alone (both $g_{ij} = 0$); **red** = both, the full rebuild. Lines are loess guides. **(a)** Largest relative RMSF error over the 98 sites. **(b)** Overlap of

the least-preserved mode among the 20 softest. **(c)** That worst-mode overlap (red) beside RMSIP over all 20 (blue), full rebuild only.

Result. The rebuild gets the *shape* of the fluctuations right and the *details* wrong.

	median over 18 states	range
RMSF profile correlation	0.982	0.961–0.995
worst-site RMSF error	22.7 %	7.6–32.9 %
RMSIP, 20 softest modes	0.973	0.956–0.995
block-5 mode overlap	0.928	0.844–0.965
max eigenvalue difference	20.1 %	up to 76.3 %
$TS_{\text{rebuild}} - TS_{\text{frustrated}}$	−4.57	−6.34 to −0.44
net change in active pairs	+25	+1 to +30

Every one is monotone in substitution number ($|\text{cor}| \approx 0.87$), so these medians are properties of the sampling grid: sampled at 10–30 the median ΔTS is −2.09, at 40–60 it is −5.53. The trend against strain in figure 5 is what carries meaning.

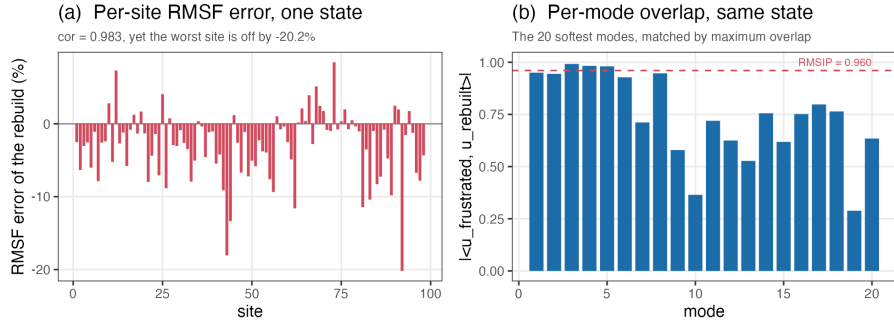


Figure 6. One state in detail — seed 501 at 60 substitutions, $E_{\text{strain}} = 36.1$, the *most strained* of the 18 and deliberately not typical. **(a)** Signed relative RMSF error at each of the 98 sites: the profile correlation is 0.983 and site 92 is still off by −20.2 %. **(b)** Overlap of each of the 20 softest modes with its assigned partner; dashed line is RMSIP = 0.960. The panel shows what a subspace score averages over. It is **not** evidence that the rebuild destroys modes — see §6.2.

What it means so far. Whole-protein measures survive the refit well — the RMSF profile correlates at 0.98, the soft-mode subspace overlaps at 0.97. Anything site-by-site does not: the worst site is off by 23 %, and the entropy shifts by several units.

That is not yet an answer, for two reasons, and the rest of §6 deals with them in turn.

- The rebuild changed *two* things at once — it dropped the transverse term **and** re-derived the contact map. Only the first is frustration. §6.1 separates them.
- An error of 23 % means nothing without knowing what a *typical* perturbation of this size would do to this spectrum. §6.2 measures that.

6.1 Is it the frustration, or the contact map?

Question. The rebuild changes two things at once: it zeroes the transverse term *and* re-derives the active set. Only the first is “neglecting frustration”. Which one does the damage?

What I did. Walk from the frustrated model to the rebuilt one in two steps instead of one, and measure how far each step moves the Hessian:

1. **Drop the transverse term**, keeping the contact map fixed. This is neglecting frustration and nothing else.
2. **Re-derive the contact map** from the coordinates, with the transverse term already gone. This is the bookkeeping change, and has nothing to do with frustration.

Size is measured as the Frobenius norm of the change in $\mathbf{K}$.

Result, first pass. Step 2 is much the larger:

	(1) frustration	(2) contact map	both	$\sqrt{(1)^2 + (2)^2}$
median over 18 states	2.53	15.09	15.31	15.38

The two add in quadrature to within 0.34 %, which is the part that could have come out otherwise: it says the two changes push $\mathbf{K}$ in nearly perpendicular directions, so neither is a disguised version of the other and comparing their sizes is meaningful. On that basis, **re-deriving the contact map does 5.6 times as much as neglecting frustration.**

That was my conclusion, and it is wrong — or rather, it is a fact about the ANM and not about the protein. Here is why.

Every contact the two models disagree about is a borderline contact. Take one state (seed 501, 40 substitutions). The two contact maps differ on 59 pairs. Every single one sits within 1 Å of the 10.5 Å cutoff, 90 % within half an ångström: the gained ones are just inside, by 0.21 Å typically, and the lost ones just outside, by 0.09 Å. Between them they hold 5.9 % of the structure’s energy.

These are the weakest springs in the protein, and the ANM step function counts each of them at **full strength** $k_{ij} = 1$. A pair at 10.4999 Å enters **K** as heavily as a tightly packed core contact. So the size of step 2 is set by how many marginal springs happen to flip, which is a property of using a step function — not a physical statement about contact maps.

Result, second pass. If that diagnosis is right, softening the cutoff should shrink step 2 and leave step 1 alone. Replace the step by a sigmoid of width w ,

$$k_{ij}(l_{ij}) = [1 + \exp((l_{ij} - d_{\max})/w)]^{-1}$$

so a borderline spring enters with a small weight instead of a full one, and apply it to **both** models so nothing else changes. It does exactly that.

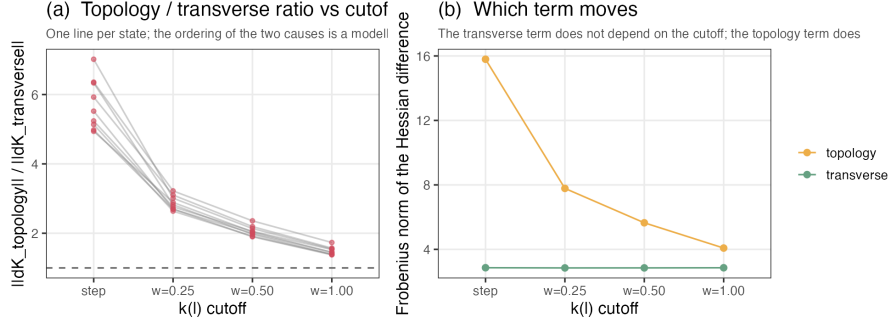


Figure 7. The same three walks sampled every 20 (9 states), re-expressed under four kernels. **(a)** One grey line per state across the four cutoffs; y is $\|\Delta\mathbf{K}_{\text{active set}}\|_F / \|\Delta\mathbf{K}_{\text{transverse}}\|_F$, dashed line at 1. **(b)** The two norms separately: transverse (green) flat by construction, active set (yellow) falling four-fold.

cutoff	transverse	active set	ratio
step	2.86	15.80	5.52
sigmoid, $w = 0.25$	2.85	7.78	2.82
sigmoid, $w = 0.50$	2.85	5.65	2.04
sigmoid, $w = 1.00$	2.86	4.08	1.45

Monotone in w in all 9 states.

What it means. *Under a hard cutoff*, re-deriving the contact map perturbs the Hessian about $5.5\times$ more than neglecting frustration does. But that factor is a property of the ANM step function, not of the protein: soften the kernel and the two become comparable. The transverse term is untouched throughout, as it must be, since it does not involve the cutoff.

The honest statement is that the two effects are of comparable size, and any claim that one dominates is a claim about the kernel.

6.2 Is the error large? A null control

Question. §6 says the worst site’s RMSF is off by 23 %. Is that a lot? The number has no meaning on its own: this protein has 288 non-rigid modes packed into a narrow range, and *any* change to $\mathbf{K}$ of comparable size will move things around. What is needed is a yardstick — how much would a perturbation of the same magnitude, but of no particular kind, disturb the same quantities?

What I did. Build that yardstick. Take the state’s own network and jitter its spring constants at random: multiply each k_{ij} by a random factor, tuning the spread until the resulting change in $\mathbf{K}$ is the same size as the rebuild’s (matched on $\max |\Delta \mathbf{K}|$ to within 5 %, by bisection).

The point is what this jitter *does not* do. It changes no rest length, so it adds and removes no frustration. It leaves the contact map untouched (checked on all 27 draws). It is a perturbation of the same magnitude as the rebuild and of no physical significance whatever — so whatever damage it does is the baseline that the rebuild’s damage has to be judged against. Three jitters per state, 9 states.

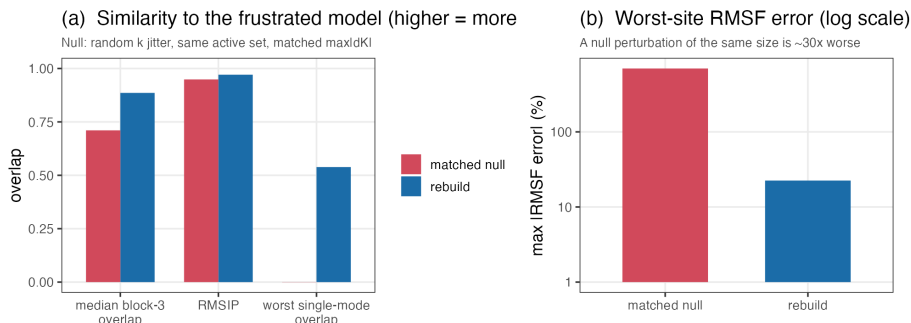


Figure 8. Rebuild (blue) against matched null (red). **(a)** Similarity to the frustrated model, higher = more similar: worst single-mode overlap, median block-3 overlap, RMSIP. **(b)** Worst-site RMSF error, log axis.

Result.

	rebuild	matched null
worst single-mode overlap	0.539	0.001
RMSIP over 20 modes	0.971	0.949
median block-3 overlap	0.886	0.710
worst-site RMSF error	22.5 %	698 %

	rebuild	matched null
max eigenvalue difference	0.221	0.997

What it means. The random jitter is worse on every measure, and on the worst-site RMSF it is worse by a factor of thirty — 698 % against the rebuild’s 22.5 %. So the honest reading of §6 inverts: **refitting is not a mildly damaging approximation, it is a remarkably faithful one.** Of all the ways you could disturb $\mathbf{K}$ by that much, discarding the frustration is among the least harmful.

It also settles what to make of the low single-mode overlaps in figure 6. The rebuild shifts eigenvalues by about 15 %, while neighbouring eigenvalues in that part of the spectrum are only about 6 % apart. When a perturbation exceeds the level spacing, eigenvectors mix within near-degenerate groups and which vector gets called “mode 10” stops being physics and starts being bookkeeping. The jitter confirms it: it drives the worst overlap to 0.001, essentially total scrambling, while barely touching RMSIP. **So nothing mode-by-mode in §6 is evidence about frustration**, and I should not have read it that way.

7. Conclusion

Frustration can be neglected, and standard practice is better justified than this exploration set out to test.

Refitting an ENM to a strained structure preserves the RMSF profile (correlation 0.98), preserves the soft-mode subspace (RMSIP 0.97, block-5 overlap 0.93), and does so far better than a random perturbation of the same magnitude would (§6.2). What it costs is site-resolved accuracy — tens of per cent at the worst sites — and a few units of TS .

Two qualifications matter more than the headline:

1. **Most of the difference is not frustration at all**, but the contact map being re-derived. Under a hard cutoff that term is $5.5\times$ the frustration term; under a soft one they are comparable (§6.1). Which dominates is a modelling choice, usually made without comment.
2. **Individual normal modes of a perturbed ENM should not be read** — in this or any model with a spectrum this dense. The null control shows the low overlaps carry no information (§6.2).

The one place the contact map clearly does matter is entropy: letting k_{ij} follow l_{ij} widens the range of $\Delta(TS)$ across sites 6.25-fold while leaving the ΔV profile essentially unchanged (correlation 0.974, §4). Adding or removing a spring changes the spectrum far more than moving a rest length does.

What would settle it. One protein, one cutoff, one k model. The obvious next step is §6.1’s decomposition under a smooth kernel throughout, where the frustration term may well become the interesting one.

Appendix A. A smooth kernel

The step makes V discontinuous in the parameters. Replacing it with the sigmoid of §6.1 and running 40 substitutions at $\nu = 1$:

	E_{strain}	s_{max} over $k > 0.5$ (Å)	acceptance
$w = 0.25$	20.97	0.810	0.571
$w = 0.50$	21.49	1.057	0.482

Not comparable with §3’s table: these ran 40 substitutions against 120, and the sigmoid gives *every* pair $k_{ij} > 0$, so the network is fully connected (4753 pairs against 960). A proper comparison needs the sigmoid thresholded so the densities match. The kernel’s effect where it does change a conclusion is reported in §6.1.

Appendix B. Verification

Three suites in `checks/`, all passing.

test_core.R — the wild-type state reproduces penm’s ANM exactly (960 edges, $\max |\Delta \mathbf{K}| = 3.6 \times 10^{-15}$ on a scale of 12.9); the minimiser reaches $|\mathbf{F}| < 10^{-8}$ with exactly 6 zero modes; ΔV splits exactly at both relaxed and strained references; $k_{ij} = k_{ij}(l_{ij})$ holds after mutation; contacts break *and* form; mutation is exactly reversible in l , k , internal structure and V .

test_profiles.R — the frustrated Hessian genuinely differs from the $g = 0$ one; six zero modes wherever a spectrum is taken; the rebuild is relaxed by construction and does change the active set; mode matching is one-to-one; every accepted trajectory state is at its own minimum.

test_can_fail.R — six deliberate sabotages, all detected. Including a numerical-Hessian arbitration of the transverse sign on a *strained* network (correct form off by 3.65×10^{-6} , flipped by 0.245), and a demonstration that off a stationary point the spectrum shows 4 zero modes and a negative eigenvalue of -4.8×10^{-4} .

Appendix C. What I got wrong

Recorded because the corrections changed conclusions, not only numbers.

The minimiser returned unconverged structures. One draw in 200 from a strained state hit the iteration limit at $|\mathbf{F}| = 2.4 \times 10^3$ and yielded $\Delta V = 1.6 \times 10^5$, poisoning a mean into 778 where every other value was 0.61. It now raises an error.

ΔV_{stress} used the wild type’s k_{ij} — the LFENM assumption $\delta k = 0$, false by construction here. It made a 26 % discrepancy look like truncation error; with k_{ij}^{mut} the gap is 0.009 %. I then “fixed” the failing check by freezing k , bending the experiment to suit an expansion this study exists to avoid.

I concluded “topology dominates frustration by 5.6 $\times$ ” without testing whether the ratio depended on the kernel. It does, completely (§6.1).

I read low mode overlaps as physical. They are not; a null control shows any perturbation of that size does worse (§6.2).

Mode matching used row-wise maxima, not a matching: 31 of 54 comparisons had two modes claiming one partner, inflating every overlap.

I quoted a block-5 overlap of 0.84 as “the defensible claim”. It is the worst of 18 states; the median is 0.93.

Smaller: a claim that all wild-type $l_{ij} \leq d_{\text{max}}$ (true only of active pairs — over all 4753 they reach 36.3 Å); a mechanism asserting active pairs “cannot drift” (falsified by 66 lost edges); an unsourced count “318 to 228” that no script produces; a transcription error in the sabotage figures.

Appendix D. Limitations

- One protein, one cutoff, one k model, one mutation size for the main result.
- $\beta = 1$ in ANM units is a convention; absolute TS values are not physical.
- The 18 states are 3 walks $\times$ 6 correlated snapshots, not 18 independent draws; effective n for trajectory-level statements is 3.
- Mutation perturbs all of a site’s pairs (**radius = Inf**), so one mutation can reach across the protein.
- Trajectories are 40–120 substitutions at a single temperature; no phylogeny.
- The penm package is not modified. The frustrated Hessian is reimplemented here because `set_enm()` blocks `frustrated = TRUE`.

Appendix E. Files

R/klfenm_core.R	state, $k(l)$, energy, Hessian, exact minimiser, mutation
R/klfenm_profiles.R	per-site and per-mode comparisons
R/klfenm_trajectory.R	scans and walks

analyses/run_all.R	§§3–6.1 and Appendix A → data/results.rds (~45 min)
analyses/run_control.R	§6.2 → data/control.rds
figures/make_figures.R	the eight figures
checks/	test_core.R, test_profiles.R, test_can_fail.R

```
Rscript analyses/run_all.R
Rscript analyses/run_control.R
Rscript figures/make_figures.R
```